



Principal Examiner's Report

January 2026

Pearson BTEC National Level 3
Applied Science/Forensic & Criminal Investigation
Unit 1: Principles and Applications of Science I
(31617H)

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Introduction

Biology

This paper is one of three externally assessed papers worth 30 marks each that contribute to Unit 1 – Principles and Applications of Science I. In this series, the paper assessed key biological concepts including the structure and function of prokaryotic and eukaryotic cells, specialised cell adaptations, Gram-staining outcomes, antibiotic action, synaptic transmission, gas exchange, and the classification of organisms using plant and animal cell organelles. Learners were required to interpret diagrams, apply biological principles to unfamiliar contexts, analyse graphical data, and construct extended comparative explanations.

Chemistry

This paper is one of three externally assessed papers, worth 30 marks each, that assess Unit 1 – Principles and Applications of Science I. This paper covers key chemistry concepts, including formulae and properties of ionic substances, dative covalent bonding, interpretation of ionisation energies, calculations using moles and the displacement reactions of halogens.

Physics

This examination was designed to allow learners to demonstrate their knowledge and understanding of 'Waves in Communication'. The topics included graphical representation of waves, harmonics, standing waves and waves of the electromagnetic spectrum. The applications included musical instruments, communication and optical fibres.

Overall Performance of the Unit

Biology

This series assessed a broad range of content across cell structure, microbiology, physiological mechanisms, data handling, and classification, with questions requiring learners to apply knowledge, interpret diagrams, explain biological processes, and construct extended comparative responses. The paper covered topics such as relating structure to function in specialised cells, interpreting staining outcomes, explaining mechanisms such as antibiotic action and synaptic transmission, analysing graphical data on oxygen saturation, and discussing classification challenges in organisms such as *Euglena*.

Overall performance across the paper was mixed, with many learners demonstrating secure recall of isolated facts but finding it more challenging to apply these in context or construct mechanistic explanations. Questions requiring straightforward identification or selection from diagrams were generally answered well, whereas items demanding precision—such as distinguishing Gram-positive from Gram-negative staining outcomes, describing ion movement across membranes, or explaining cooperative binding—revealed gaps in learners’ deeper conceptual understanding. Questions with the command verb of explain, particularly those requiring multi-step reasoning, were the most discriminating. Many learners tended to describe observable features or restate information rather than explain underlying mechanisms, which limited access to higher-level marks.

Misconceptions were a recurring barrier, including confusion between respiratory and cardiovascular conditions, confusion of prokaryotic and eukaryotic features, and inaccurate statements about organelle distribution. Extended responses often contained irrelevant or incorrect information, and imprecise language—such as referring to organelles as cells or organisms—was common. In several questions, exam technique also played a role, with learners repeating information from the stem, overlooking command words such as “explain,” or failing to provide the comparative reasoning required for higher levels.

Despite these challenges, many learners showed emerging strengths in interpreting trends, identifying key structures, and recognising general biological principles. Where candidates engaged with the question requirements and applied their knowledge logically, they were able to access higher marks. However, the overall pattern suggests that learners need to continue to develop their ability to articulate mechanistic explanations, use precise scientific terminology, and apply knowledge flexibly across unfamiliar contexts.

Chemistry

Learners that were successful this year showed a good understanding of key terms, scientific vocabulary and mathematical skills and were able to apply this knowledge and understanding to new scenarios. They were able to use correct scientific conventions when writing formula and equations. They engaged with the questions well and used the prompts in the questions to inform their answers and applied their answers to the context given rather than giving generic answers. They were precise with their language, were able to link ideas and explain in detail where necessary. Learners that performed well, were able to explain trends in ionisation energy across a period and interpret graphs of ionisation energy to link these to the electronic structure

of an element. They were able to carry out moles calculations in a clear and logical manner, showing their working at all points and explain the order of reactivity of the halogens by using the outcomes of some of their displacement reactions.

Physics

Most learners were able to correctly identify the amplitude of a wave and determine its wavelength from a graphical representation. The majority of learners could perform at least some aspects of calculations using the equations provided in the questions.

Learners were generally able to suggest how string tension would affect the frequency of a wave on the string, but many struggled to explain the link between the two.

Learners who performed well were able to correctly complete a calculation which involved rearranging a square root term and could describe how standing waves are produced, with correct use of appropriate technical terminology.

Although the topic of optical fibres has appeared in many of the recent examination series, a large number of learners are still struggling to explain how their design leads to total internal reflection of a pulse of light.

Most learners were able to discuss the use of radio waves and microwaves in satellite communication, but only a few were able to include knowledge with the information provided in the question and provide the highest level of response.

Individual Questions

Biology

Question 1

Q1(a) - learners generally performed well on this item, with many correctly identifying the outer membrane from the diagram. Responses suggested that most learners were confident in recognising the key structural features of a bacterial cell when presented visually. Some learners confused the outer membrane with the cell membrane. Overall, this question successfully differentiated between learners with secure knowledge of bacterial cell organisation and those with more superficial recall about cells.

Q1(b) - performance on this question was mixed, with only around half of learners selecting the correct colour. A substantial proportion of incorrect

responses indicated purple or purple/pink combinations, suggesting that many learners defaulted to the Gram-positive outcome rather than applying their understanding to the context provided. There was also a notable number of references to colourless or green, which do not relate to the Gram-staining process and indicate gaps in learners' procedural knowledge.

The following response was awarded 1 mark. Pink or red are creditworthy colours. We can accept "shades" of pink/red if they are seen.

(b) State the colour a Gram-negative bacterium would be after Gram staining.

(1)

pink

Q1(c) – learners generally demonstrated secure understanding in this question, with most correctly identifying the nucleoid. This suggests that the concept of the nucleoid is well established for the majority of learners.

Q1(d) – learner performance on this extended-response item was variable. Many learners successfully gained the first marking point by recognising that antibiotics kill bacteria, although fewer explicitly referred to slowing or inhibiting bacterial growth. This indicates a broadly secure understanding of the general purpose of antibiotics, even if the mechanistic detail was sometimes limited. For the second and third marking points, some learners were able to describe disruption to a bacterial structure, most commonly the cell wall. However, many responses lacked precision, with vague statements such as antibiotics “going through” or “penetrating” the membrane, which did not demonstrate understanding of a specific structural or functional disruption. A notable proportion of learners also showed confusion with staining techniques, incorrectly suggesting that antibiotics would cause bacteria to change colour or turn purple. Others incorrectly proposed that antibiotics activate or support white blood cells, reflecting misconceptions about the distinction between antimicrobial agents and immune responses.

The following response was awarded 2 marks. The learner inhibiting the division of cell wall is sufficient for MP2. The ideas of destroying/killing/ slowing down growth/ reduce reproduction/multiply of the bacteria are all acceptable for MP1. Therefore the learner can be awarded MP1 for their comment about "can't reproduce".

An antibiotic can be used to treat an infection caused by Gram-positive bacteria.

(d) Explain the effect the antibiotic has on the Gram-positive bacteria.

(2)

Antibiotics ~~the~~ inhibits the division of cell wall of bacteria so new bacteria daughter cells can't be produced and the bacteria can't reproduce and multiply

(Total for Question 1 = 5 marks)

The following response was awarded 1 mark. The learner has not given MP1 as there is no reference to destroying the bacteria. However the marking points are independent from each other and so there is sufficient here for MP2 – as the learner has stated that the antibiotic would break down the peptidoglycan.

An antibiotic can be used to treat an infection caused by Gram-positive bacteria.

(d) Explain the effect the antibiotic has on the Gram-positive bacteria.

(2)

The antibiotic would break down the peptidoglycan layers of the gram positive bacteria. gram positive bacteria is sensitive to antibiotics

(Total for Question 1 = 5 marks)

Question 2

Q2(a) – learners generally approached this calculation with confidence, and many were able to complete the required initial step, whether using a difference-first method or direct substitution into the percentage-increase formula. Both approaches were seen frequently, and most learners demonstrated awareness that the calculation required multiplication by 100 to convert to a percentage. However, errors in substituting values were common and represented the main barrier to achieving full marks. These mistakes often occurred despite learners showing a clear understanding of the overall method, suggesting that accuracy rather than conceptual misunderstanding was the limiting factor. Where learners progressed to the evaluation stage, many rounded their answers appropriately.

The following response was awarded 3 marks. The learner has given 2 different

methods. However, they have selected the method that gave the answer 473.3 as they have written this on the answer line and therefore that is the answer we mark.

We cannot mix and match marks from the two methods – so you must either credit marks for method 1 or for method 2, so for example, 573.3 and 86-15 are marking points from the two different methods and so the learner could only be given 1 mark in total, not two, as we award marks up to 3 for the steps within one method. The correct answer has been given on the answer line. If there is no answer on the answer line, then we look at the value at the end of the learners working. As the answer is correct 473.3 (with or without the recurring dot) then we can award the 3 marks, without investigating the working.

There does not have to be working, but by giving working ECF or process marks can be given if the incorrect answer is given, but we can give benefit of the doubt (BOD) that the working was completed correctly when the correct answer is seen.

We can accept 473.3333 (with or without the recurring dot), 473.3 or 473.

We cannot accept 470 as this is insufficient significant figures.

2 A human brain contains 86 billion neurones.

The peripheral nervous system connected to the brain contains 15 billion neurones.

- (a) Calculate the percentage increase in the number of neurones in the brain compared to the peripheral nervous system.

(3)

Show your working.

$$\frac{86}{15} \times 100$$

$$= 573.\dot{3}$$

$$\frac{86 - 15}{15} \times 100 = 473.\dot{3}\%$$

$$\text{percentage increase} = 473.\dot{3}\%$$

The following response was awarded 1 mark. In this example we can see MP1 of method 1 which is the difference of $86 - 15$. Even without an evaluation this can be awarded MP1. Either $86 - 15$ or 71 can be awarded MP1 as we can presume that they have completed this substitution (This is only 1 mark as there is not an individual evaluation mark for this stage of the calculation).

The learner has then carried out some calculations and has given 17 as their final answer. There is nothing creditworthy after the 71.

2 A human brain contains 86 billion neurones.

The peripheral nervous system connected to the brain contains 15 billion neurones.

(a) Calculate the percentage increase in the number of neurones in the brain compared to the peripheral nervous system.

(3)

Show your working.

$$86 - 15 = 71$$

$$71 \div$$

$$100 - 86 = 14$$

$$100 - 15 = 85$$

percentage increase = 71 17 %

Q2(b) - performance on this multi-mark question was mixed. Many learners successfully accessed the first marking point by recognising that sodium channels open when the neurotransmitter binds to receptors on the post-synaptic membrane. However, a proportion of responses were negated by references to channels "opening and closing," which did not accurately reflect the required mechanism. There was also evidence of confusion between ligand-gated and voltage-gated sodium channels, although this was only penalised when learners incorrectly stated that sodium ions themselves opened the channel. The second marking point was awarded less often, with many learners failing to specify clearly that sodium ions move into the post-synaptic neurone. Ambiguous descriptions of ion movement, or statements that did not identify the destination of the ions, limited credit. The final marking point was more commonly achieved, with a greater number of learners correctly referring

to depolarisation, even if they did not fully articulate the associated change in membrane potential leading to threshold. Overall, responses showed partial understanding of synaptic mechanisms, but precision in describing ion movement and channel function was often lacking.

The following response was awarded 2 marks. The learner has given MP1 – when the sodium channel opens – please note that the context of neurotransmitters is not required – providing that they do not negate this – for example, if they said "the sodium causes the channel to open then MP1 could not be awarded. They have almost given MP2 as they have said when the sodium ions flow in, and later they have said the sodium goes across the synaptic cleft TO the post synaptic neurone – but the sodium has not actually entered the neurone and so we cannot award MP2. We don't need to hold out for ions or the +, sodium or Na is sufficient.

The learner has stated that the sodium ions increase the voltage and that this is depolarisation – this is sufficient for MP3 in line with the additional guidance. Just a reference to increasing the voltage does not fully answer the question they must state increasing the voltage to reach the threshold/to a value between – 60mV and +45mV (we do not need to hold out for the unit)/to cause depolarisation.

Figure 2 shows a synapse between two neurones.

Sodium channels are found on the post-synaptic membrane.

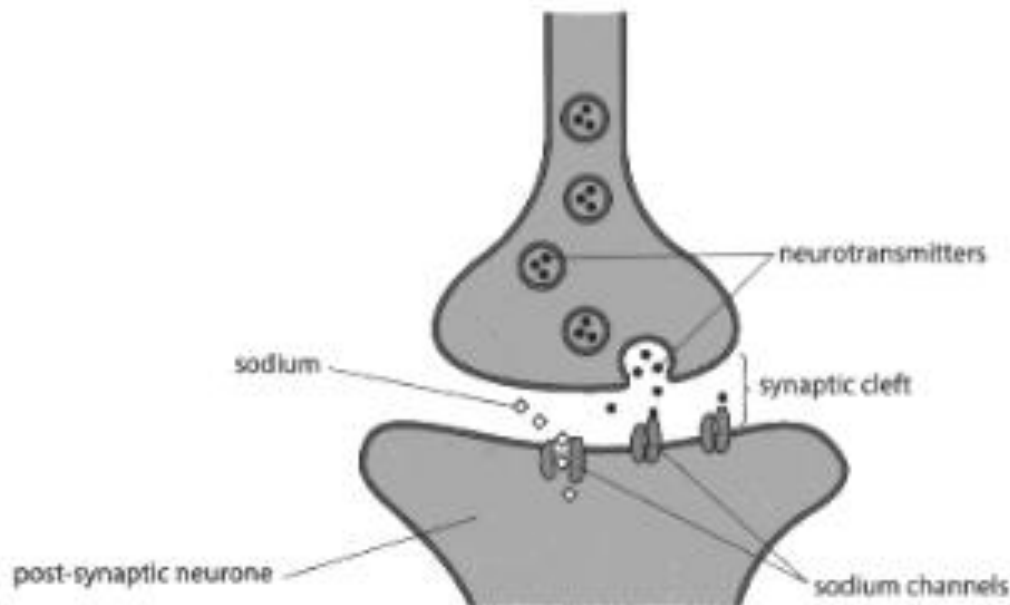


Figure 2

(b) Describe the role of the sodium channels in triggering an action potential.

(3)

When the sodium channel opens Na^+ ions flow in with a positive charge.
The Na^+ ions increase the voltage.
This process is called depolarization.
When the sodium channel gate closes it will have reached its peak which is called the action potential.
(Total for Question 2 = 6 marks)
Sodium goes across the synaptic cleft from the pre-synaptic neurone to the post-synaptic neurone. Where they trigger another set of neurotransmitters up an action potential.

The following response was awarded 1 mark. The learner has given enough for MP3 using values within the range in the additional guidance of the mark scheme. The marking points can be given in isolation and the comments about the sodium channels diffusing can be ignored.

Figure 2 shows a synapse between two neurones.

Sodium channels are found on the post-synaptic membrane.

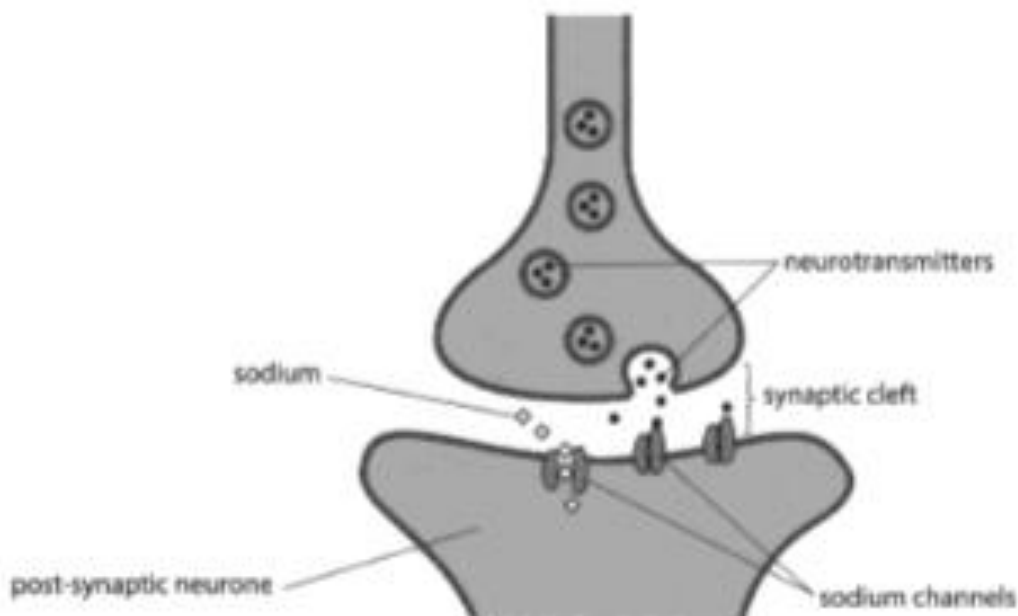


Figure 2

(b) Describe the role of the sodium channels in triggering an action potential.

(3)

When an action potential is triggered, the threshold increases beyond -55mV and triggers the release of sodium channels. Sodium channels diffuse into making the inside of the post-synaptic membrane more positive, and the outside more negative after $+30\text{mV}$ sodium channels close quickly.

(Total for Question 2 = 6 marks)

Question 3

Q3(a) – learners generally found this question challenging, and overall performance was weak. A large number of responses incorrectly referred to red or white blood cells. Many learners also identified ciliated or goblet cells, suggesting that they defaulted to more familiar examples of epithelial tissue rather than interpreting the context of the question. Even where learners recognised that the answer related to epithelial tissue, many simply stated “epithelial cell” without specifying squamous epithelium, which was required for credit.

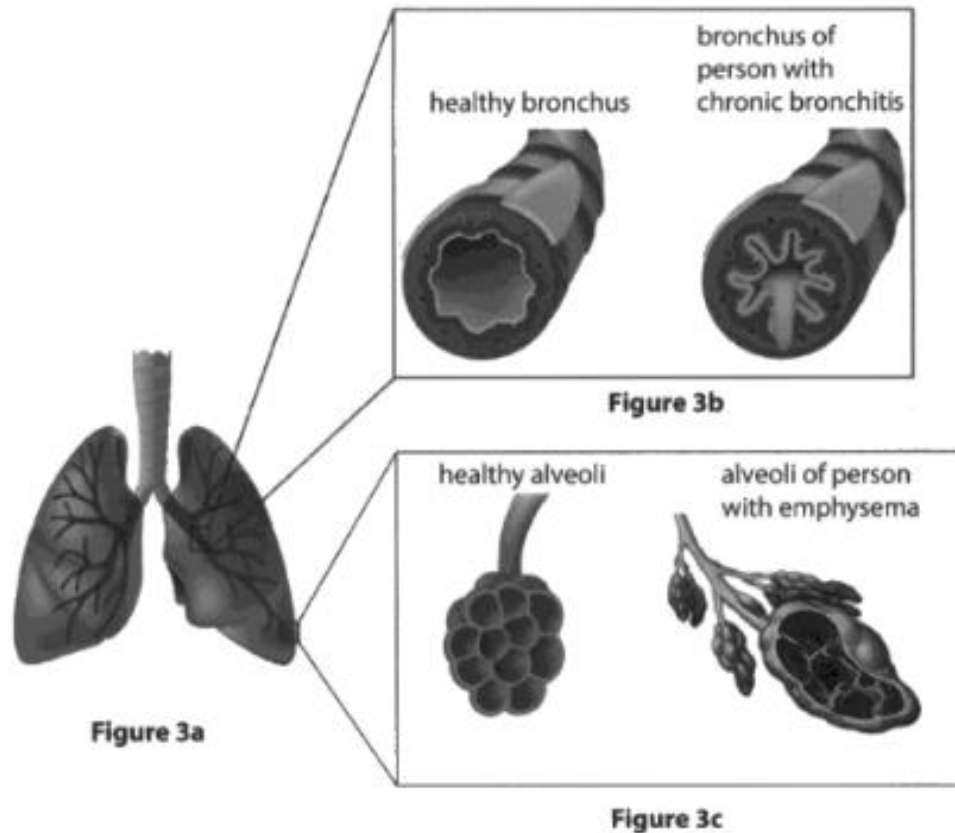
The following response was awarded 1 mark. This is creditworthy – we are crediting the idea of squamous and so we can accept squamous epithelial, squamous epithelium or just squamous. However, endothelium would negate the mark, so “squamous endothelium” would be 0 marks.

Some responses referring to “pavement epithelium” have been seen and we would credit this with the mark – please add this to your mark scheme but ignore pavement cells or pavement unqualified.

3 Figure 3a shows a diagram of the lungs.

Figure 3b shows a healthy bronchus and the bronchus of a patient with chronic bronchitis.

Figure 3c shows healthy alveoli and the alveoli of a patient with emphysema.



[Source: CPAL]

(a) Name the type of cells that line the alveoli.

(1)

Squamous epithelial cells

Q3(b) – this question was answered relatively well, with many learners correctly referring to symptoms such as breathlessness, coughing, and wheezing. These responses demonstrated a secure understanding of the respiratory symptoms associated with bronchitis. However, a substantial number of learners confused the respiratory system with the cardiovascular system, giving answers related to plaque build-up, restricted blood flow, or high blood pressure—indicating confusion with atherosclerosis. Additionally, many responses mentioned “mucus” without specifying excessive or increased mucus production, which limited access to full credit. Some learners also confused symptoms with causes, and incorrectly gave lifestyle factors, rather than symptoms.

The following response was awarded 2 marks. The learner has given a list of 3 possible answers:

- difficulty breathing
- chest pain
- cough

All 3 are listed on the NHS website as symptoms of chronic bronchitis and so the maximum 2 marks can be awarded. If there was an incorrect symptom on their list, then the list principle would apply.

(b) State **two** symptoms of chronic bronchitis. (2)

1 Difficulty in exhaling and inhaling.

2 Pain in the lungs/coughing attacks.

Q3(c) – learners generally struggled with this multi-stage explanation, and very few achieved more than one mark. Most learners were able to recognise that gas exchange was reduced, but responses often lacked the precision required to access the first marking point. Many referred vaguely to “damaged alveoli” without explicitly linking this damage to a decreased surface area for gas exchange. This omission prevented learners from demonstrating the causal mechanism required for full credit.

The second marking point was the most commonly achieved, with many learners correctly stating that less oxygen enters the bloodstream, muscles, or cells. However, a significant number of responses remained too vague, stating only that there was “less oxygen in the body” or incorrectly suggesting that less oxygen entered the lungs.

For the third marking point, although many learners mentioned respiration, they rarely specified aerobic respiration. This lack of precision meant that the link between reduced oxygen availability and reduced aerobic respiration was not clearly established. References to anaerobic respiration were also common but irrelevant in this context.

The final marking point was awarded least frequently. Very few learners made the explicit connection between reduced aerobic respiration and reduced ATP production. Instead, many described general symptoms such as tiredness or breathlessness, which did not demonstrate understanding of the underlying physiological mechanism. Overall, responses tended to describe effects rather

than explain the biological pathway, revealing gaps in learners' ability to construct multi-step mechanistic explanations.

The following response was awarded 2 marks. The learner has not given enough for MP1 as there is no reference of reduced surface area- they need both reduced surface area with a link to gas exchange for MP1. They have just given enough for MP2 with their comment about haemoglobin isn't binding to a lot of oxygen to transport around body cells- the cells comment is just enough and the link to haemoglobin implies the oxygen is in the blood. They have referenced respiration, but there is no mention of aerobic and so MP3 cannot be awarded. They have then mentioned less ATP and so MP4 can be given. The learner must imply that there is less ATP or energy, not just that ATP is needed for exercise.

(c) Explain why a person with emphysema could find it difficult to exercise.

(4)

• A person with emphysema can find it difficult to breathe because the alveoli ~~in there~~ are damaged, this means ^{and carbon dioxide} the rate of diffusion of oxygen is lower than it should be, this means the body won't be getting enough oxygen to the heart, slowing down the heart rate and this causes a shortness of breath.

• Haemoglobin isn't binding to a lot of oxygen to transport around body cells for respiration, this means they won't have enough energy

(Total for Question 3 = 7 marks)

because oxygen is needed for respiration and if there's not enough oxygen, the mitochondria won't make a sufficient amount of ATP needed to exercise.

The following response was awarded 1 mark. The learner has stated reduces the surface area for gas exchange, and so MP1 can be awarded. They have commented that less oxygen is going into the body, which is too vague for MP2. Any comments about carbon dioxide can be ignored. They have stated lack of oxygen needed for respiration, but we must hold out for the word aerobic (this is in bold in the mark scheme and so much be stated for the mark point).

We can ignore references to anaerobic respiration.

(c) Explain why a person with emphysema could find it difficult to exercise.

(4)

emphysema is a disease caused by inhaling harmful substances and chemicals like ~~cigarette~~ smoking.

- These harmful chemicals breakdown the alveoli
- This reduces the surface area for gas exchange
- This means less oxygen is going into the body and less carbon dioxide is being released out of the body.
- This means when exercising there is a real lack of oxygen needed for respiration
- You will have to use anaerobic respiration but this may cause cramps because of lack of ~~for~~ oxygen.

(Total for Question 3 = 7 marks)

Question 4

Q4(a) – performance on this question was mixed, with many learners losing credit due to exam-technique issues rather than gaps in biological understanding. A significant number of learners repeated information already provided in the question stem, which could not be credited. Among those who did offer an alternative explanation, the most common correct response referred to increased flexibility, enabling red blood cells to pass more easily through narrow capillaries. Overall, the question highlighted the importance of careful reading and interpretation, as many learners appeared to understand the biology but did not apply it appropriately.

The following response was awarded 1 mark. The learner has described the red blood cell as platelet shape – which could mean plate, but this comment can be

ignored. They have then said allowing it to move flexibly – which is sufficient for MP1.

Reference to change of shape alone is too vague for MP1 – we need the idea of flexible or bendy.

Their comment of through the arteries is not enough for MP2 – There needs to be an idea of passing or squeezing through a narrow vessel, narrow artery narrow vein, narrow lumen, or capillary (without a description we can give benefit of the doubt for a capillary being narrow), but just fitting through a space is insufficient for MP2. We can accept "small" for narrow.

4 Figure 4 shows a diagram of a red blood cell.

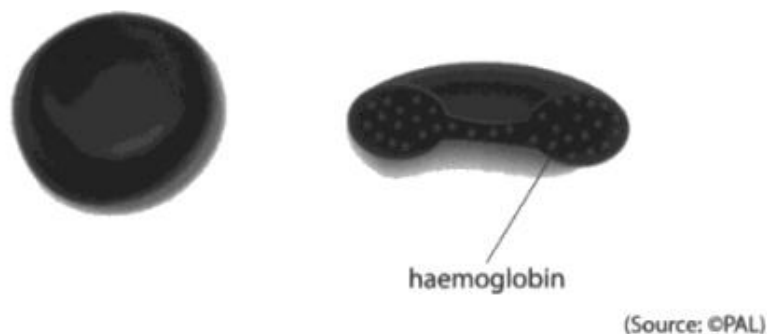


Figure 4

Red blood cells are able to contain a lot of haemoglobin because they do not have a nucleus.

(a) Explain **one other** advantage of the lack of a nucleus in a red blood cell.

(2)

The red blood cell ~~is~~ has a platelet shape, allowing it to move flexibly through the arteries.

↳ Red blood cells can bend/change shape to fit through blood networks.

Q4(b) – required learners not only to extract key information from the figure—such as the positive trend between partial pressure and haemoglobin saturation and the presence of four oxygen-binding sites—but also to apply logical biological reasoning to explain how the ease of oxygen binding changes as sites become occupied. Most learners limited their responses to a description of the

curve's shape, without providing the mechanistic explanation needed to demonstrate higher-level analytical understanding."

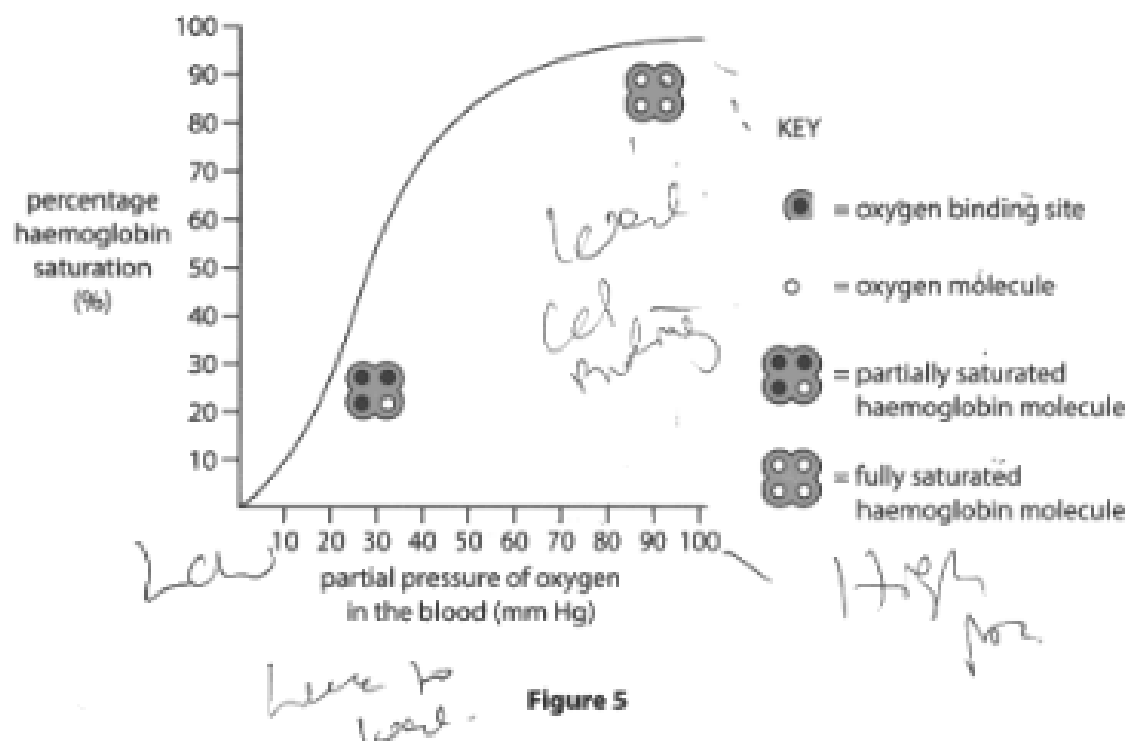
Most learners were able to access the first marking point by correctly stating that haemoglobin saturation increases as the partial pressure of oxygen increases. Many responses appropriately referenced the overall trend shown in the graph, demonstrating secure interpretation of the data. However, a common limitation was misinterpretation of the command word "explain." Rather than explaining the mechanism behind the pattern, many learners simply described the shape of the curve, noting changes in gradient without linking these to oxygen binding behaviour.

Only a minority of learners referred to haemoglobin's four oxygen-binding sites or recognised that the initial binding of oxygen is difficult at low partial pressures. References to conformational change following the first oxygen molecule binding were rare, and very few learners articulated the concept of cooperative binding—that oxygen binding becomes progressively easier as haemoglobin changes shape, before becoming more difficult again as sites become saturated. As a result, fully developed mechanistic explanations were uncommon. Most responses demonstrated partial understanding of the trend but lacked the depth and precision required to explain the underlying biochemical processes.

The following response was awarded 3 marks. The learner had given us the idea that at low partial pressures of oxygen it is harder for oxygen to bind - this is MP3. They have also said that at high partial pressures the oxygen will load readily to the haem group - giving the idea of co-operative binding - this is sufficient for MP4. The learner has also stated that increasing partial pressure of oxygen makes it easier for the haemoglobin be to saturated and this is sufficient for MP1 as learner speak. This is a borderline description for MP1.

Figure 5 shows the percentage saturation of haemoglobin with oxygen at different partial pressures of oxygen.

The saturation of haemoglobin increases as it binds oxygen.



(b) Explain the effect of increasing partial pressure of oxygen on haemoglobin saturation.

Use information from Figure 5 to support your answer.

10-20%

(4)

At a Low partial pressures of O_2 , it is harder for O_2 to bind to the haem groups. At High partial pressures of O_2 , O_2 will load readily to the haem groups (competitive binding). So, as you increase the O_2 partial pressure, it will become easier for the haemoglobin to be saturated.

(Total for Question 4 = 6 marks)

The following response was awarded 2 marks. The learner has given the idea that a low oxygen pressure the percentage haemoglobin saturation is less - this is just sufficient for MP3. The learner has stated that as oxygen pressure increases between 20 and 50mm Hg the haemoglobin saturation rapidly increases- this is sufficient for MP1. We can take the learners respond "holistically" and pair up their statements to allow for the award of MP1. However be careful that the trend is the correct way round and that the percentage haemoglobin saturation is not causing an increase in partial pressure of oxygen. Although the learner has highlighted the 20-50 mmHg section of the curve and stated there is a rapid increase, this is not enough for MP5 as there needs to be the explanation or link between the small increase of partial pressure causes a large/rapid increase of saturation.

The learner has also given the idea that haemoglobin saturation slows as it becomes fully saturated - but this is too descriptive and not enough as an explanation to award MP6.

Figure 5 shows the percentage saturation of haemoglobin with oxygen at different partial pressures of oxygen.

The saturation of haemoglobin increases as it binds oxygen.

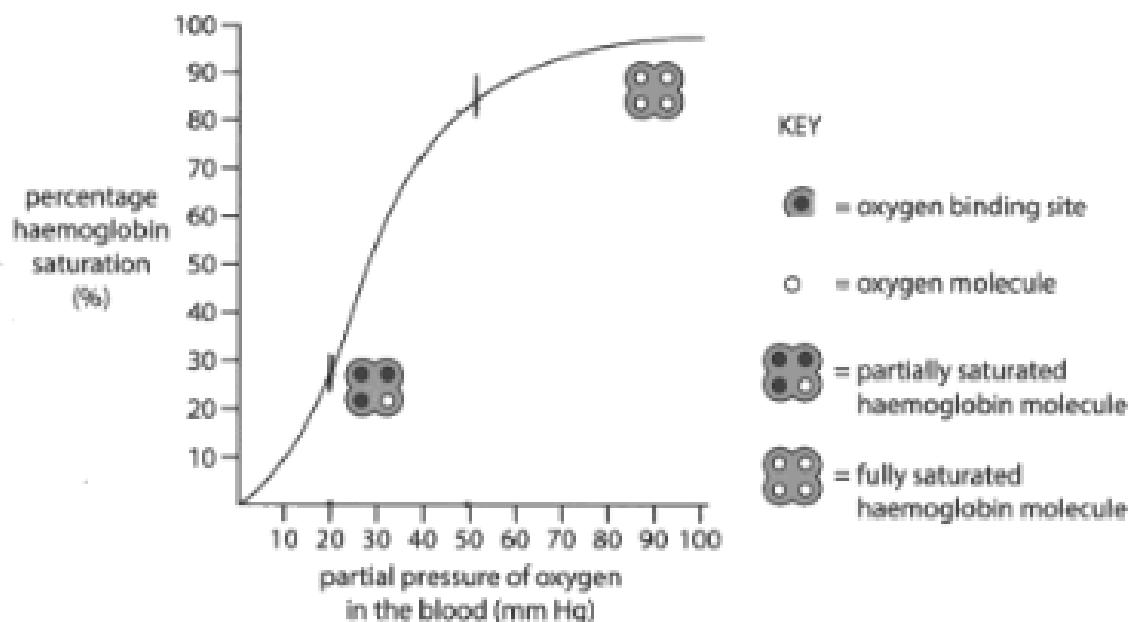


Figure 5

(b) Explain the effect of increasing partial pressure of oxygen on haemoglobin saturation.

Use information from Figure 5 to support your answer.

(4)

When there is a low oxygen pressure haemoglobin is less saturated since it has less oxygen to bind to. However as oxygen pressure increases between 20 and 50 mm Hg haemoglobin saturation rapidly increases from 25 to 80 per cent due to an influx of oxygen to bind to. Finally as oxygen pressure increases from 50 to 100 mm Hg haemoglobin saturation slows as it starts to become fully saturated between 80-100 mm Hg.

(Total for Question 4 = 6 marks)

Question 5

Overall, most learners produced responses that met the criteria for Level 2, typically achieving a mark of 4 where they included some relevant scientific detail and demonstrated an element of discussion rather than simple description. Many learners were able to make at least one comparative statement and offer some explanation of why *Euglena* is difficult to classify, which was sufficient to lift their response into Level 2. However, progression to Level 3 was uncommon. The main barrier was the inclusion of substantial irrelevant or incorrect information, alongside a lack of sustained comparative reasoning needed to support a fully developed discussion. A high number of learners made inaccurate claims about organelle distribution—for example, stating that Golgi apparatus or mitochondria are found only in animal cells, or that contractile vacuoles occur only in plant cells. Surprisingly, many responses also asserted that only animal cells contain a nucleus. Some learners listed organelle functions without linking these features to specific cell types, while others used imprecise language, referring to organelles as cells or organisms. Incorrect references to sperm cells having flagella were also common and so did not contribute to the discussion. These misconceptions, combined with limited comparative structure, prevented many learners from demonstrating the depth and accuracy required for Level 3.

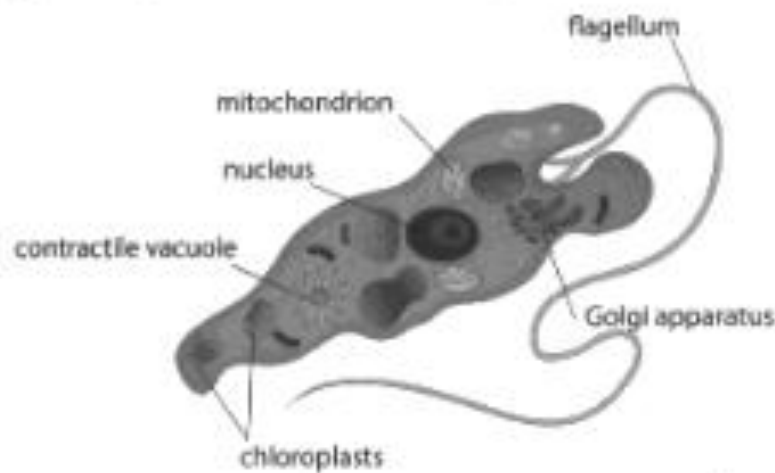
Level 3: Higher-level responses included most of the relevant scientific content and demonstrated clear evaluation. These candidates explained why *Euglena* cannot easily be classified as strictly plant or animal, referencing features such as photosynthesis due to chloroplasts alongside motility (as in a sperm cell) and the absence of a cell wall. Stronger responses explicitly evaluated the classification dilemma, justifying why it may not fully fit either category. Fully developed, well-balanced discussions were less common.

The following response was awarded 5 marks, level 3. The learner has stated that the prokaryote does not have a nucleus but could have a flagellum. They have stated that it could be eukaryotic because it contains mitochondria, nucleus and Golgi apparatus. They have mentioned that a plant cell contains chloroplasts and a vacuole (but they have not considered the contractile element of vacuole would have pushed this closer to being 6 marks).

There is a brief mention of being an animal cell, but they have not used the features to say could be an animal cell - if they had said about absence of cell wall this would have been sufficient to award 6 marks for a fully comprehensive discussion information about each organelle has made the discussion disjointed - we need to pick out what is relevant to the question - we should ignore information about the structure and function of organelles as this does not answer the question, although is good knowledge, the presence/absence of cells to discuss how to classify *Euglena* is the focus on the question.

Figure 6 shows a Euglena.

Euglena is a single cell organism that is difficult to classify.



(Source: DPAL)

Figure 6

Discuss, using the information in Figure 6, why Euglena is difficult to classify.

(6)

You should include:

- what type of cell Euglena can be characterised as
- reference to both plant and animal cell characteristics.

Euglena can be characterised as a eukaryotic cell as it contains features of both plant and animal cells. Such as mitochondria which has a double membrane. The inner membrane is tightly folded to form cristae and the central part is called the matrix. The function of the mitochondria is where cellular respiration occurs. It also has a Golgi apparatus which structure is a stack of membrane bound flattened sacs and its function is that it receives proteins from rough ER, modifies them and sends them back to where they are needed. It also contains a nucleus which is not a feature in

prokaryotic cells like bacteria despite Euglena having
 a flagellum which is a feature in prokaryotic cells not
 eukaryotic cells like animals and plants. The nucleus
 has a double membrane and is the largest organelle
 surrounded by a nuclear envelope which allows the
 movement of molecules to pass through it. The function
 of the nucleus is to control cellular activity and contains
 genetic material such as DNA. However, it has a flagella
 (a coil) which allows the cell to move to nutrients and
 harmful materials. Euglena also contains features of
 a plant cell as it has chloroplasts which has a double
 membrane and is filled with a fluid called stroma.
 The inner membrane consists of a flattened network of
 cells called thylakoids. Its function is to trap light
 energy using chlorophyll for photosynthesis. Also has
 a vacuole which is a membrane bound sac found in
 cell membrane filled with cell sap. Molecules outside to
 ensure rigid framework of cell. Therefore, can be
 classified as either prokaryotic or eukaryotic cell.

Level 2: The majority of candidates achieved this level. Responses included relevant scientific knowledge, typically recognising that Euglena is a eukaryote, supported by reference to the presence of a nucleus and membrane-bound organelles. Many answers considered both plant and animal characteristics, such as the presence of chloroplasts (supporting classification as a plant cell) and the absence of a cell wall (supporting classification as an animal cell). Where candidates structured their response to compare these features and made some attempt at discussion, Level 2 marks were awarded.

The following response was awarded 3 marks, level 2. The learner has included a common incorrect idea and suggested that Euglena could be an animal cell as it has flagellum, and considers the sperm cell. This idea can be ignored. Incorrect

or irrelevant comments can be ignored and then we consider what is left to decide on a level and mark, unless this contradicts – for example the learner states that chloroplasts are needed for a cell to photosynthesis suggesting this is a plant or an animal cell. Therefore credit for the plant cell having a vacuole has been negated by the contradiction that the animal cell also has chloroplasts.

The learner continues and says that it would be difficult to characterise it as a plant cell because it had a vacuole, nucleus...and chloroplasts so this is a little muddled and not clear how these ideas are supporting the discussion, later in the responses. However from the bottom of the first page and the information on the second page the learner has given some creditworthy information – the learner has considered that the cell cannot be a prokaryote because it has a nucleus and then has correctly expressed their idea about the plant cell vacuole. Therefore there is *just* enough to place this at the bottom of level 2 and award 3 marks.

5 Figure 6 shows a Euglena.

Euglena is a single cell organism that is difficult to classify.

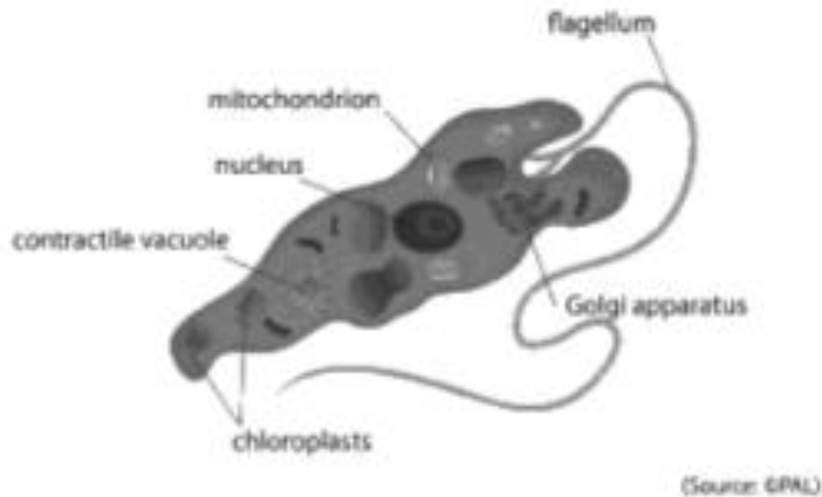


Figure 6

Discuss, using the information in Figure 6, why Euglena is difficult to classify.

(6)

You should include:

- what type of cell Euglena can be characterised as
- reference to both plant and animal cell characteristics.

A Euglena can be characterised as is difficult to be ~~not~~ characterised as a sperm cell as it has a flagellum for movement, but it has a nucleus instead of a haploid nucleus which a sperm cell requires as it makes it easier to move. It would be difficult to characterise it as a plant cell as it has a contractile vacuole, nucleus, mitochondrion and chloroplasts, while it has a flagellum, which plant cells do not require ~~these are~~. However chloroplasts are need for a cell to photosynthesise suggesting this is a plant or animal cell. Euglena is not a prokaryote as it has a

nucleus whereas prokaryotes don't have a nucleus and their genetic material is loose. However it also indicates that it is not a plant cell as plant cells have a large vacuole for structure instead of a contractile vacuole.

Level 1: Responses at this level were characterised by limited scientific detail and were often presented as lists of features with little or no development. Some candidates identified isolated plant or animal characteristics without linking these to classification or explaining why this creates difficulty.

The following response was awarded 2 marks, level 1. The learner has given ideas about the organism being a bacterium, but they have incorrectly stated that the prokaryote has a nucleus and this is a contradiction.

They have correctly stated that a prokaryote could have a flagellum.

They have briefly considered the chloroplasts, which are usually found in plants – but this line of explanation is cut short.

They have then suggested that the animal body (which is a little confused as this could mean they think the Euglena has an animal body, or they mean cell) has a vacuole, Golgi apparatus and mitochondria – but this suggests incorrectly that a plant cell does not have these and so this is a confused line of support for the discussion. There seems to be a misconception that only animal cells have Golgi apparatus.

Overall this is a level 1, 2 marks, as there is a beginning of a discussion but there are contradictions throughout.

5 Figure 6 shows a Euglena.

Euglena is a single cell organism that is difficult to classify.

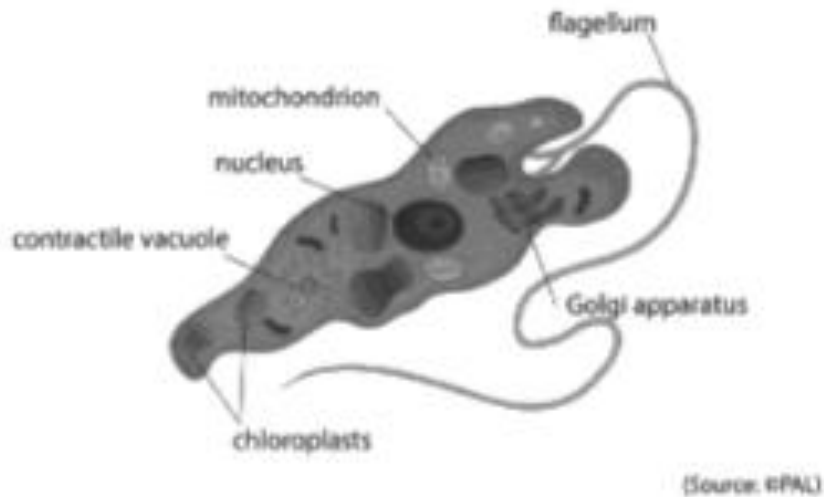


Figure 6

Discuss, using the information in Figure 6, why Euglena is difficult to classify.

(6)

You should include:

- what type of cell Euglena can be characterised as
- reference to both plant and animal cell characteristics.

Forget Euglena can be ~~characterised~~ characterised as a ^{prokaryote} ~~eukaryote~~ or a pathogen/bacteria. This is because it has a nucleus, a flagellum, flagellum and ~~contractile vacuole~~ ~~vacuole~~ ~~no~~ ~~has~~ ~~no~~ ~~cap~~ ~~has~~ a vacuole. This type of bacteria would be considered as a good bacteria due to it's benefits within the ~~bottom~~ ~~organism~~ ~~the~~ ~~animal~~ ~~body~~ ~~and~~ ~~plants~~. This bacteria has a flagellum, meaning that it is able to freely move around using the ~~mitochondria~~ energy from it's mitochondria.

Although, it is difficult to ~~disagree~~
~~characterise~~ characterise* due to it
 having chloroplasts. Chloroplasts are usually
 found in plants for photosynthesis.
~~This~~ It also has a vacuole and
 a golgi apparatus* which is found
 in the animal body.* + mitochondria.

Chemistry

Question 1

The first question on the paper asked learners about the ionic compound, potassium nitrate.

Q1(a)(i) gave learners the formula of the potassium ion and the nitrate ion, learners were then asked to write the formula of the potassium nitrate. Learners found this difficult with many losing marks as they did not remove the charges on the ions from the overall compound, as in this example which did not score.

1 (a) Potassium nitrate is an ionic compound that is used in some fertilisers.

Potassium nitrate contains potassium ions, K^+ and nitrate ions, NO_3^- .

(i) State the formula of potassium nitrate.

(1)

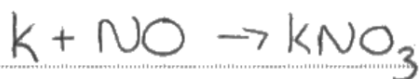


Some did not know what was meant by the term formula and tried to write a balanced equation instead of the formula of the potassium nitrate. This did not score.

1 (a) Potassium nitrate is an ionic compound that is used in some fertilisers.

Potassium nitrate contains potassium ions, K^+ and nitrate ions, NO_3^- .

(i) State the formula of potassium nitrate.



(1)

In some cases, learners did not score as they were not accurate when representing the symbols in the formula, giving lower case letters rather than upper case and using a superscript for the 3 rather than a subscript 3. Learners should be taught to write formula using the correct scientific conventions.

This example shows a good answer which scored the mark.

1 (a) Potassium nitrate is an ionic compound that is used in some fertilisers.

Potassium nitrate contains potassium ions, K^+ and nitrate ions, NO_3^- .

(i) State the formula of potassium nitrate.



(1)

Q1(a)(ii) and (a)(iii) were multiple-choice questions.

Q1(a)(ii) – learners were required to apply their understanding of the properties of ionic compounds and use it to identify the melting point of potassium nitrate. In general learners performed well in this question, with a good proportion knowing that the melting point would be high as it is an ionic substance and therefore selecting D as their answer.

Q1(a)(iii) – learners were asked which of four types of substance would not conduct electricity. Learners found this more challenging than the previous multiple-choice question with fewer understanding that solid potassium nitrate would not conduct electricity and selecting D as their answer.

Q1(a)(iv) asked learners to explain how molten potassium nitrate conducts electricity. The majority of learners performed poorly on this question because they did not read the question carefully and thought that the question asked how potassium rather than potassium nitrate conducts electricity. With the most common answer relating to delocalised electrons being able to move, rather than ions moving to carry the charge. This example scored no marks.

(iv) Explain how molten potassium nitrate conducts electricity.

(2)

When potassium nitrate is molten the electrons ~~are~~ are no longer tightly packed and free to move therefore carry electronic charge in order to conduct electricity

The following example scored 1 mark. The learner knows that the ions can move in the molten potassium nitrate but makes no reference to the ions carrying the charge or current to gain the second mark.

(iv) Explain how molten potassium nitrate conducts electricity.

(2)

When potassium nitrate is molten, the ions begin to move / flow over each other. This means that the electricity is then able to flow through the molten KNO_3

This last example scored shows a good example that scored both marks.

(iv) Explain how molten potassium nitrate conducts electricity.

(2)

Molten potassium nitrate conducts electricity because its ions are free to move. When potassium nitrate becomes molten it dissociates into separate ions which move and carry electric charge.

Q1(b) asked learners about a different ionic compound, ammonium nitrate. They were asked specifically about the ammonium ion, shown a diagram of the ion and asked how the dative covalent bond in the ion was formed. Learners found it very difficult to apply their understanding of dative covalent bonding with only the strongest gaining the full three marks available. Some learners were able to give a generic definition of a dative covalent bond, but fewer were able to relate this specifically to the ion, showing an

understanding of which atom would be sharing two electrons and which element was initially electron deficient.

The following example scored no marks. There is an idea that a pair of electrons comes from one atom, but there is no idea that the electrons are shared between the two atoms. There is no explanation of the hydrogen ion being electron deficient or it being the species with the empty orbital. They do not state that the pair of electrons come from the nitrogen atom.

Ammonium nitrate is another ionic compound that is used in some fertilisers.

Figure 1 shows a diagram of an ammonium ion.

The ammonium ion contains covalent bonds.

One of these bonds is a dative covalent bond.

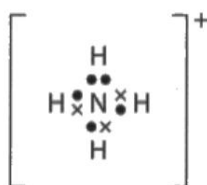


Figure 1

(b) Explain how the dative covalent bond in Figure 1 is formed.

(3)

A dative covalent bond is formed when one atom donates a lone pair of electrons to another atom that has an empty orbital. The atom donating the pair is the donor and the one receiving is the acceptor.

This second example scored 2 marks. The learner has understood that the bond is made up of a shared pair of electrons and that both of the electrons come from the nitrogen. The answer does not explain that the hydrogen ion does not have any electrons or is electron deficient to gain the last mark.

Ammonium nitrate is another ionic compound that is used in some fertilisers.

Figure 1 shows a diagram of an ammonium ion.

The ammonium ion contains covalent bonds.

One of these bonds is a dative covalent bond.

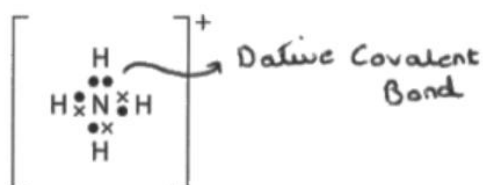


Figure 1

(b) Explain how the dative covalent bond in Figure 1 is formed.

(3)

In ammonium nitrate the lone pair of electrons from the nitrogen is shared with the hydrogen atom to complete its outer shell. This is known as a dative covalent bond.

This last example shows a good response that scored all 3 marks.

Ammonium nitrate is another ionic compound that is used in some fertilisers.

Figure 1 shows a diagram of an ammonium ion.

The ammonium ion contains covalent bonds.

One of these bonds is a dative covalent bond.

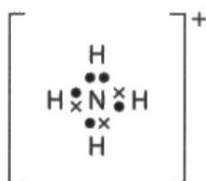


Figure 1

(b) Explain how the dative covalent bond in Figure 1 is formed.

(3)

A dative covalent bond is when both shared electrons in a covalent bond are from the same ^{atom} ~~atom~~, this occurs in Figure 1 because one of the Hydrogen atoms is a positively charged ion that has no electrons thus meaning that both the electrons it shares comes from the Nitrogen atom.

Question 2

The second question on the paper asked about the elements carbon and nitrogen.

Q2(a)(i) told learners that nitrogen contains 7 electrons and asked them to complete an electron-in-boxes diagram to show the arrangement of electrons in the atom. In general learners performed well in the question with a good proportion being able to represent the electrons as full or half arrows, placing them in the correct orbitals and understanding that where an orbital contained two electrons, the arrows needed to be in opposite directions to gain both marks. In this example, the learner has used full arrows, half arrows were also accepted.

2 (a) Carbon and nitrogen are in the same period of the periodic table.

A nitrogen atom contains 7 electrons.

(i) Complete Figure 2 to show the arrangement of electrons in a nitrogen atom.

(2)

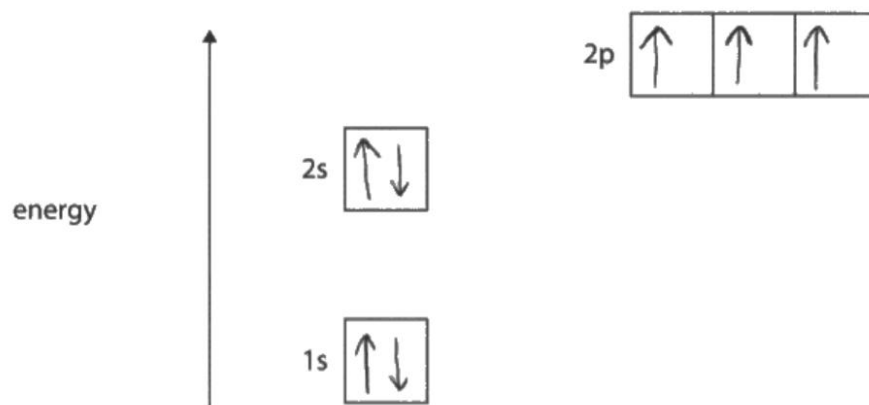


Figure 2

Where learners did not score full marks, it was often as they did not understand that the orbitals should be filled singularly before pairing, as in this example. The learner has understood that where there are two electrons in an orbital, that they should be of opposite spin to gain the second marking point, but they did not recall that the three 2p electrons should be single electrons, as in this example which scored 1 mark.

- 2 (a) Carbon and nitrogen are in the same period of the periodic table.

A nitrogen atom contains 7 electrons.

- (i) Complete Figure 2 to show the arrangement of electrons in a nitrogen atom.

(2)

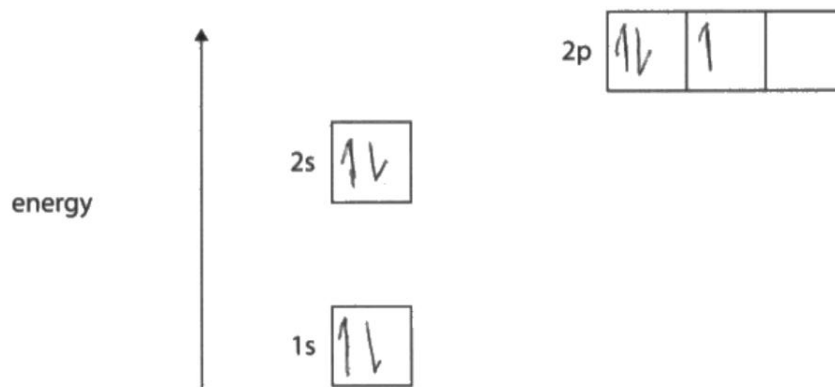


Figure 2

In some cases, it was evident that learners were not familiar with electron-in-boxes diagrams as instead of attempting to draw electrons using arrows, half or full, they simply put numbers into the boxes. This was not accepted and so this examples such as this gained no marks.

- 2 (a) Carbon and nitrogen are in the same period of the periodic table.

A nitrogen atom contains 7 electrons.

- (i) Complete Figure 2 to show the arrangement of electrons in a nitrogen atom.

(2)

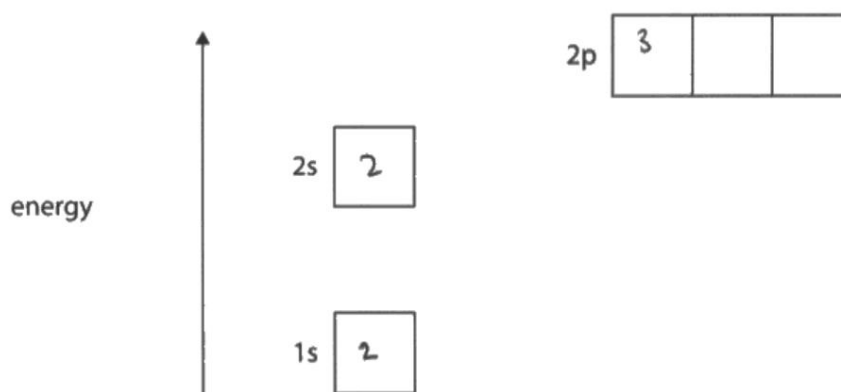


Figure 2

Q2(a)(ii) gave learners a table showing the first ionisation energy of carbon and the first ionisation energy of nitrogen. They were asked to explain why the first ionisation energy of nitrogen is greater than the first ionisation energy of carbon. Learners found it difficult to explain accurately that due to the nitrogen having more protons or a greater nuclear charge, this would mean that the electron in the outer shell would be more strongly attracted to the nucleus or the protons. Some learners discussed differences in atomic radius, this was ignored. However, a noticeable proportion of learners thought that the number of shells of electrons decreased between nitrogen and carbon which explained the difference. This incorrect idea was rejected.

Many learners stated that the amount of energy to remove the electron in nitrogen would be greater, but this was given in the stem of the question and so gained no further credit. Learners should be taught not to repeat information from the stem of the question as this will gain no credit.

This example scored 1 mark for the understanding that nitrogen has more protons than carbon. The comment regarding the atomic radius was ignored.

Table 1 shows the first ionisation energy of carbon and of nitrogen.

element	first ionisation energy (kJ mol^{-1})
carbon	1086
nitrogen	1402

Table 1

- (ii) Explain why the first ionisation energy of nitrogen is greater than the first ionisation energy of carbon.

(2)

~~The nitrogen has~~ Nitrogen has one more proton than carbon and the same amount of shells as carbon, meaning that nitrogen's atomic radius is bigger than carbon's atomic radius.

scored no marks. This example scored no marks as they thought that the element was in group 2 and period 2.

Figure 3 shows a graph of successive ionisation energies of an element, X.

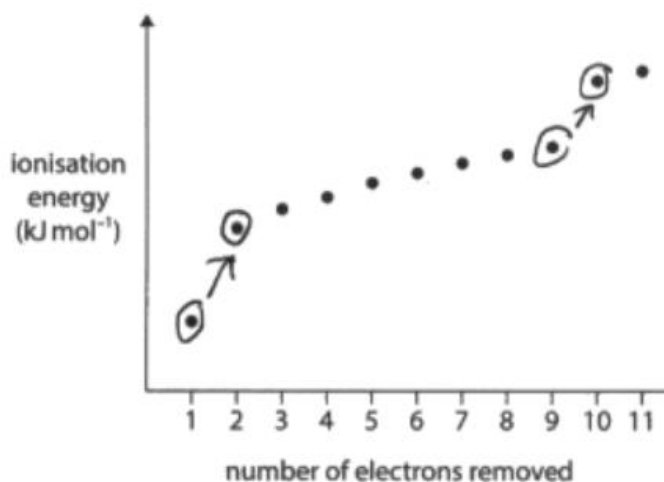


Figure 3

(b) Explain in which group and in which period of the periodic table element X can be found.

(4)

element X can be found in the 2nd
group of the periodic table due to
its rising on the chart when two electrons
are removed. Element X can be found
in the ^{2nd} period.

(Total for Question 2 = 8 marks)

This example scored 2 marks, the learner was able to recognise that the element could be found in group 1 and period 3 but were not able to explain this in terms of the ionisation energies shown in the graph and so did not gain the explanatory mark points.

Figure 3 shows a graph of successive ionisation energies of an element, X.

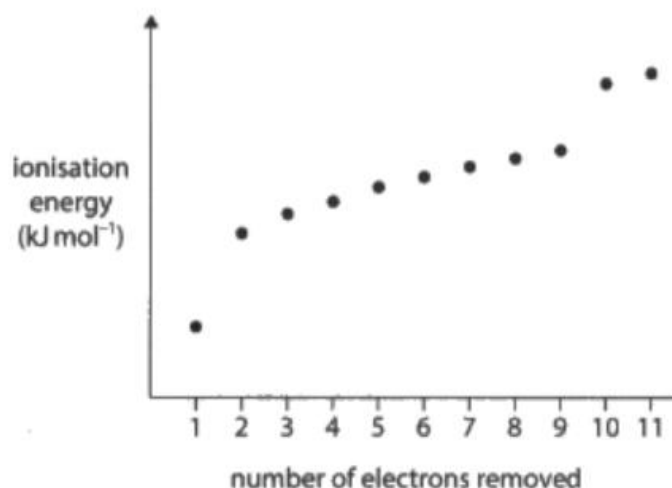


Figure 3

(b) Explain in which group and in which period of the periodic table element X can be found.

(4)

Analysing figure 3, the ionisation energy required to remove one electron was low, meaning element X most likely had 1 outer electron as it is easy to remove. The ionisation energy gradually increased as more electrons were removed with the last two requiring a lot of energy. Counting how many electrons were removed, 11 were removed which if put in an electron configuration would be 2, 8, 1. This means it has three shells, putting it in period 3 and Group 1 as it has 1 outer electron.

This example shows a good answer which explained, using the information in the graph, which group and period of the periodic table the element could be found.

Figure 3 shows a graph of successive ionisation energies of an element, X.

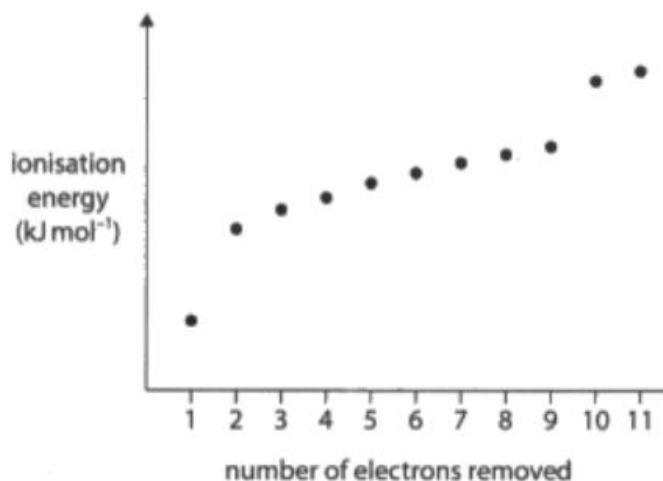


Figure 3

(b) Explain in which group and in which period of the periodic table element X can be found.

(4)

There are 2 large jumps/gaps in ionisation energy, so element X has 3 shells, so is in period 3 of the periodic table.

There is a gap/jump in ionisation energy after one electron ^{is removed}, so element X has one electron on its outer shell, so is in group 1 of the periodic table.

Element X has 11 electrons, so is Na. The electron configuration of element X is $1s^2 2s^2 2p^6 3s^1$.

(Total for Question 2 = 8 marks)

Question 3

Q3 focused on the transition metal iron.

Q3(a) asked learners to state one property of transition metals. In general, learners performed well in the question with the majority being able to give a

correct property of transition metals, as in this example which gained the mark.

3 Iron is a transition metal.

(a) State **one** property of transition metals.

(1)

high melting point ✓

Some learners stated that a property of transition metals is that they can be used as catalysts as exemplified by this answer. Whilst this is a correct use, it is not a property and so did not score the mark. Learners should be taught key terms such as property and use so that they do not lose marks for confusing the two.

3 Iron is a transition metal.

(a) State **one** property of transition metals.

(1)

can be used as catalysts.

In other cases, learners did not read the question carefully and tried to give the name of a transition metal rather than a property or other uses, both examples scored no marks.

3 Iron is a transition metal.

(a) State **one** property of transition metals.

(1)

Sodium / Na

3 Iron is a transition metal.

(a) State **one** property of transition metals.

(1)

Jewellery horseshoes (the for horses)

Q3(b)(i) told learners about the reaction of iron with steam to form an iron oxide and a gas. They were given a partially completed equation and asked to complete the state symbols for the iron and the steam and the formula for the gas produced. Learners often scored 1 mark for this question for either writing both state symbols correctly or for giving the correct formula for hydrogen, two marks were seen less often.

It was pleasing to see that a good proportion of learners knew that hydrogen was a diatomic molecule and were able to represent this correctly giving the symbol as H₂ rather than H alone, H² or 2H.

Where learners lost a mark for the state symbols, it was often as they did not read the question carefully and thought that water was either a liquid or in aqueous form. The majority were able to give the state symbol of iron as (s) correctly.

This example scored 1 mark for the correct formula of hydrogen. The mark for the state symbols was lost as they gave the symbol of steam as (l) rather than (g).

(b) Iron reacts with steam to form an iron oxide and a gas.

(i) Complete the chemical equation for the reaction by:

- adding the state symbol for iron and the state symbol for steam
- adding the formula for the gas produced.

(2)



In some cases, learners gave the words for the state symbols rather than the initials. Whilst capitals were accepted, words were not accepted and this example scored 1 mark only for the formula of hydrogen.

(b) Iron reacts with steam to form an iron oxide and a gas.

(i) Complete the chemical equation for the reaction by:

- adding the state symbol for iron and the state symbol for steam
- adding the formula for the gas produced.

(2)



This answer shows a good example which scored both marks.

(b) Iron reacts with steam to form an iron oxide and a gas.

(i) Complete the chemical equation for the reaction by:

- adding the state symbol for iron and the state symbol for steam
- adding the formula for the gas produced.

(2)



Q3(b)(ii) gave learners the actual yield of a reaction of some iron with steam and the theoretical yield and then asked learners to identify the percentage yield of iron oxide in the reaction. Learners performed well in the question with most learners being able to select the correct percentage yield as 65% to gain the mark.

Q3(b)(iii) – learners were asked to calculate the number of moles in 11.57g of the iron oxide. The relative formula mass of the iron oxide was given. Learners performed very well in the question with the majority of learners gaining the mark.

(iii) Calculate the number of moles in 11.57 g of the iron oxide.

(relative formula mass: $\text{Fe}_3\text{O}_4 = 231.4$)

$$\text{moles} = \frac{M}{M_r}$$

$$\text{moles} = \frac{11.57}{231.4} \quad (1)$$

0.05 ←

number of moles of iron oxide = 0.05

Where learners did not score, it was often as they got the sum upside down to gain an answer of 20 or multiplied the answer. Whilst this did not score in (b)(iii), error carried forward was accepted in (b)(iv) and so full marks could still be scored in (b)(iv) if the working was shown and correct.

Q3(b)(iv) – learners found it very hard to calculate the number of iron ions in the 11.57g of iron oxide. Some who knew that they needed to multiply the number of moles by Avogadro's constant scored two marks but often they forgot to multiply by three to obtain the number moles of ions in the iron oxide.

This example scored 2 marks as they have multiplied the number of moles obtained in (b)(iii) by Avogadro's constant.

(iv) Calculate the number of ions of iron present in 11.57 g of iron oxide.

(Avogadro's constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

(3)

Show your working.

If you did not get an answer for part (b)(iii), use 0.04 for the number of moles of iron oxide.

↗

$$\underline{0.05} \times (6.02 \times 10^{23}) =$$
$$3.01 \times 10^{22}$$

number of iron ions = 3.01×10^{22}

In this example, the learner has also given an answer of 3.01×10^{22} . However, as they have shown no working, no marks can be awarded. This example highlights the importance of learners doing as asked in the stem of the question and showing their working.

(iv) Calculate the number of ions of iron present in 11.57 g of iron oxide.

(Avogadro's constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

(3)

Show your working.

If you did not get an answer for part (b)(iii), use 0.04 for the number of moles of iron oxide.

number of iron ions = 3.01×10^{22}

(Total for Question 3 = 8 marks)

The next example scored 0 marks in part (b)(iii) and 1 mark in part (b)(iv). In part (iii) although they have the correct calculation, they have not evaluated it correctly and so the mark was not awarded. However, error carried forward was allowed into (b)(iv). In (b)(iv) the learner has not multiplied the number of moles by three and so they did not gain the first mark. With error carried forward they did score the second mark for multiplying their number of moles by Avogadro's constant. Unfortunately, they have then carried out another incorrect step so they did not score the evaluation mark.

(iii) Calculate the number of moles in 11.57 g of the iron oxide.

(relative formula mass: $\text{Fe}_3\text{O}_4 = 231.4$)

$$\frac{11.57}{231.4} = 0.049 \text{ moles mass ratio (1)}$$

number of moles of iron oxide = 0.049

(iv) Calculate the number of ions of iron present in 11.57 g of iron oxide.

(Avogadro's constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

(3)

Show your working.

If you did not get an answer for part (b)(iii), use 0.04 for the number of moles of iron oxide.

$(6.02 \times 10^{23}) \times 0.049$
 $= 2.9498 \times 10^{22}$
 $\frac{2.9498 \times 10^{22} \times 11.57}{11.57} = 34.129186$
 $= 2.549$
 number of iron ions = 34.13

number of iron ions = 34.13

This example shows a good answer; they have shown that the number of ions in the formula is three and so have multiplied the number of moles by three. They have then multiplied this by Avogadro's constant and so scored all three marks.

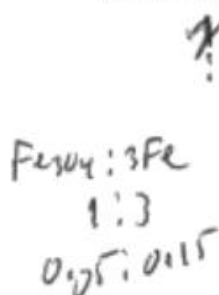
(iv) Calculate the number of ions of iron present in 11.57 g of iron oxide.

(Avogadro's constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

(3) ✖

Show your working.

If you did not get an answer for part (b)(iii), use 0.04 for the number of moles of iron oxide.



$$\begin{aligned}
 &\cancel{0.05 \times (6.02 \times 10^{23})} = \cancel{3.01 \times 10^{22}} \\
 &0.15 \times (6.02 \times 10^{23}) = 9.03 \times 10^{22}
 \end{aligned}$$

$$\text{number of iron ions} = \frac{9.03 \times 10^{22}}{\cancel{3.01 \times 10^{22}}}$$

Question 4

The last question on the paper was the six mark question with a level-based mark scheme. Learners were given some information about reactions of the halogens. They were given two reactions, the reaction of sodium chloride with bromine water and the reaction of sodium iodide with bromine water. They were told that the first reaction did not react and that the second reaction was a redox reaction. They were then asked how the experiment gives evidence of the order of reactivity of the three halogens bromine, chlorine and iodine and asked them to include the order of reactivity of the three halogens, the expected observations of from the experiment, an equation for the redox reaction and why the reaction is a redox reaction.

Learners seemed to access the question well with the majority attempting the question. However, the vast majority thought the reactivity of the halogens increased rather than decreased down the group, thinking that iodine would be the most reactive and chlorine the least. Many tried to explain the increase or decrease in reactivity based on electronic configuration which was often correct but not relevant to the question and so did not gain any credit. They then try to explain the reactions based on the incorrect order of reactivity which often confused their answer. Learners that did try to explain why the reaction was a redox reaction often got the species that were

oxidised and reduced the wrong way around. Only the strongest learners were able to give the oxidation and reduction in terms of oxidation states of each of the species and gave the correct colour changes involved.

Unfortunately, in this example, although the learner gives a correct order of reactivity to start, this is contradicted later in the answer, this was commonly seen. The learner has a few basic, unlinked ideas which were creditworthy as bromine gaining electrons and an attempt at an equation to show the displacement, this was just sufficient for 2 marks in level 1.

4 Bromine, chlorine and iodine are three halogens.

An experiment is carried out to determine the order of reactivity of these three halogens.

A sample of sodium chloride solution and a sample of sodium iodide solution are placed in separate test tubes.

A few drops of orange bromine water are added to each test tube and the tubes are shaken.

Figure 4 shows the equipment used.



Figure 4

There is no reaction when bromine water is added to sodium chloride solution.

There is a redox reaction when bromine water is added to sodium iodide solution.

Explain how this experiment gives evidence for the order of reactivity of the three halogens.

Your answer should include:

- the order of reactivity of the three halogens
- the expected observations from the experiment
- an equation for the redox reaction
- why the reaction is a redox reaction.

OIL RIG

Reactivity in halogens decreases as you go down the group. Meaning bromine is most reactive than chlorine than iodine.

(6)

- As you go down the group reactivity decreases as there are more shells making you further from the nucleus. This means iodine is most reactive then chlorine then bromine.

- Redox is when electrons are lost and gained electrons. In this case bromine is ~~reduced~~ ^{reduced and} and sodium iodide is ^{lost} ~~gained~~ an electron.



As you go down the group reactivity decreases. As you get further from the nucleus as there is more shells. This means the order of reactivity is chlorine then bromine then iodine.

(Total for Question 4 = 6 marks)

In this example, the learner gives the correct order of reactivity and links this to the reason why bromine cannot displace the chlorine in the sodium chloride. They say that bromine displaced the 'iodine' and give a correct word equation to show the displacement. They say that is redox because one halogen is being reduced and one is being oxidised. Although they do not say which one did which, this does show that they understand redox. There is no explanation in terms of oxidation and reduction in terms of oxidation states or correct species losing/gaining electrons.

This answer was just sufficient to gain full credit in level 2 and 4 marks were awarded.

4 Bromine, chlorine and iodine are three halogens.

An experiment is carried out to determine the order of reactivity of these three halogens.

A sample of sodium chloride solution and a sample of sodium iodide solution are placed in separate test tubes.

A few drops of orange bromine water are added to each test tube and the tubes are shaken.

Figure 4 shows the equipment used.

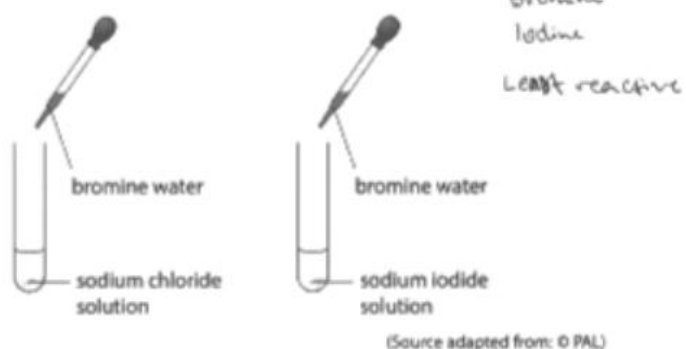


Figure 4

There is no reaction when bromine water is added to sodium chloride solution.

There is a redox reaction when bromine water is added to sodium iodide solution.

Explain how this experiment gives evidence for the order of reactivity of the three halogens.

Your answer should include:

- the order of reactivity of the three halogens ☒
- the expected observations from the experiment ☒
- an equation for the redox reaction ☒
- why the reaction is a redox reaction. ☒

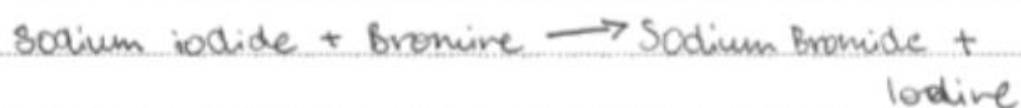
(6)

Firstly, the bromine solution did not react or displace the chlorine in the sodium chloride solution because chlorine is more reactive than bromine.

◦ Secondly, when the bromine water was added to the sodium iodide, the bromine displaced the iodine forming sodium bromide and iodine (redox).

◦ Therefore, we can conclude that the halogens in group 7 are arranged in order of reactivity down the group.

◦ The redox reaction of sodium iodide + Bromine



◦ This reaction is a redox reaction because one halogen element is being reduced and one halogen is being oxidised, displacing one another.

◦ The expected observation from the experiment would be when bromine water was added to sodium chloride no reaction would occur whereas when added to sodium iodide the solution would turn orange and reaction would occur.

(Total for Question 4 = 6 marks)

This last example shows an answer which scored the full 6 marks available. The learner has covered all four ideas in their answer, the correct order of reactivity, the expected observations, an equation for the redox reaction and why the reaction is a redox reaction in terms of oxidation states. An excellent answer which had more than sufficient for full credit in level 3.

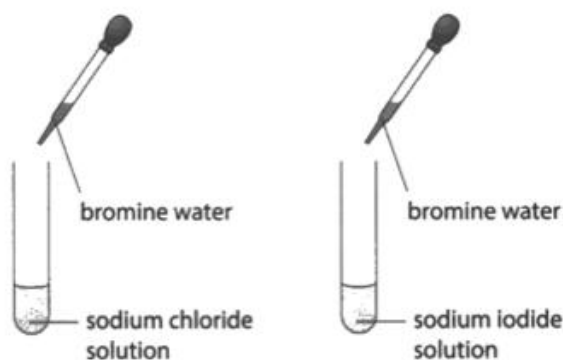
4 Bromine, chlorine and iodine are three halogens.

An experiment is carried out to determine the order of reactivity of these three halogens.

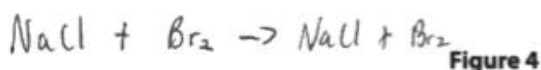
A sample of sodium chloride solution and a sample of sodium iodide solution are placed in separate test tubes.

A few drops of orange bromine water are added to each test tube and the tubes are shaken.

Figure 4 shows the equipment used.



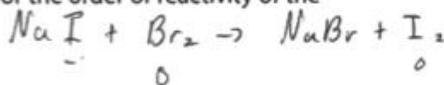
(Source adapted from: © PAL)



There is no reaction when bromine water is added to sodium chloride solution.

There is a redox reaction when bromine water is added to sodium iodide solution.

Explain how this experiment gives evidence for the order of reactivity of the three halogens.



Your answer should include:

- the order of reactivity of the three halogens
- the expected observations from the experiment
- an equation for the redox reaction
- why the reaction is a redox reaction.

(6)

In these three halogens reactivity decreases down the group, so chlorine is the most reactive and Iodine is the least reactive. This is because halogen need to gain one more electron for a full outer shell, so Chlorine has the least amount

halogen need
to gain one
electron
chlorine
most reactive

P
S
C
M
A
Z
I
C

Feasible
 $\text{Mg} + \text{K}_2 \rightarrow$

of electron shells so its outer electron is more attracted to the nucleus and is closer so it takes more energy to lose the outer electron making it the most reactive.

Observations for bromine ^{water} and sodium chloride will be that the orange bromine ~~in~~ water will stay the same colour.

Observations for bromine water and sodium iodide will show a dark brown solution.

The redox reaction will be $2\text{NaI} + \text{Br}_2 \rightarrow 2\text{NaBr} + \text{I}_2$

The reaction is a redox reaction as Iodine oxidation number goes from -1 to 0 showing oxidation and Bromine oxidation number will go from 0 to -1 showing reduction. Reduction and oxidation are happening in the same experiment.

Physics

Question 1

Q1(a)(i) – learners were provided with the diagram shown below and asked to identify the amplitude of the wave.

- 1 (a) Figure 1 shows a displacement–distance graph for a wave.

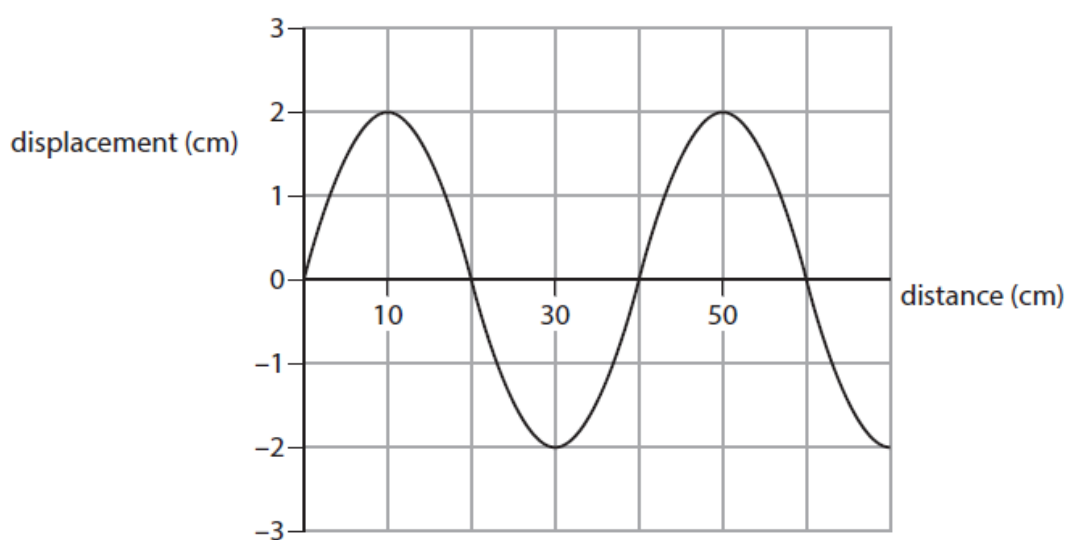


Figure 1

- (i) Identify the amplitude of the wave in Figure 1.

(1)

- ☐ A 2 cm
- ☐ B 4 cm
- ☐ C 20 cm
- ☐ D 40 cm

The majority of learners correctly selected option A, 2 cm.

Q1(a)(ii) – from the same diagram learners were asked to determine the wavelength of the wave. This proved more challenging, with just over half of learners able to correctly determine the wavelength as 40 cm.

- (ii) Determine the wavelength of the wave in Figure 1.

(1)

wavelength = 40 cm

Q1(b) – for this question learners had to analyse and interpret a diagram showing two waves and identify the phase difference between the waves.

Figure 2 shows a displacement–distance graph for two waves.

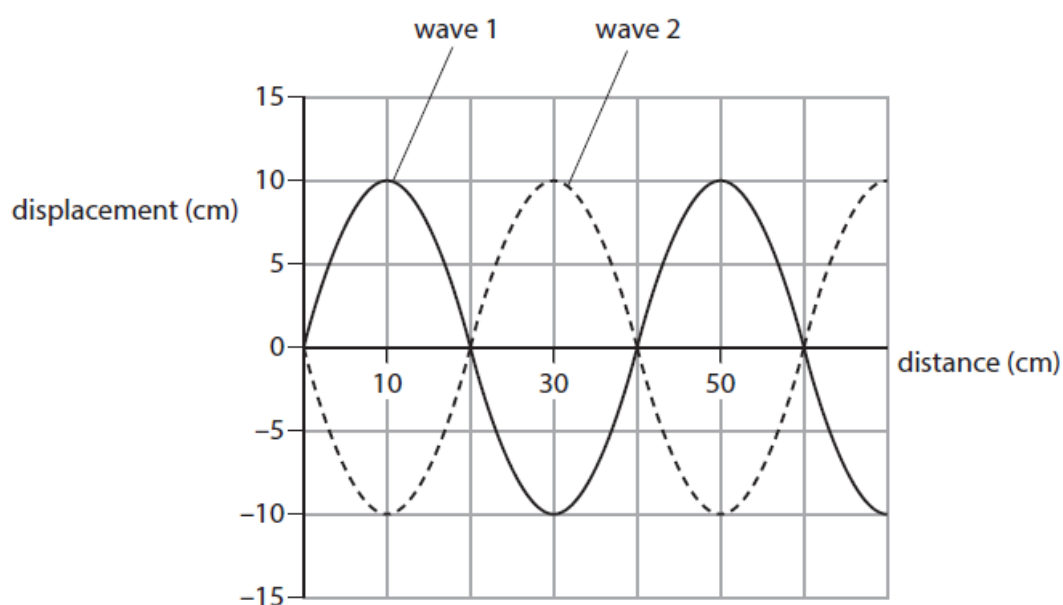


Figure 2

(b) Identify the phase difference between wave 1 and wave 2 in Figure 2.

(1)

- ☐ A 45°
- ☐ B 90°
- ☐ C 180°
- ☐ D 270°

Most learners correctly identified the phase difference as 180° , option C.

Q1(c) – few learners were able to correctly calculate the frequency of the wave in this question, with the most common mistake being either incorrectly converting the time to seconds or missing this step out altogether. The following response gained full marks.

Use the equation

$$f = \frac{1}{T}$$

Show your working.

$$f = \frac{1}{4.0 \text{ ms}} = 0.25$$

$$4.0 \text{ ms} \div 1000 = 0.004 \text{ s}$$

$$f = \frac{1}{0.004 \text{ s}} = 250$$

frequency = 250 Hz

Centres should encourage learners to show their working when completing calculations, as many can gain marks for the process used, even if their final answer is incorrect. The response below is an example of this.

Use the equation

$$f = \frac{1}{T}$$

$$4 \text{ ms} = 4000$$

Show your working.

$$f = \frac{1}{4000} = 25 \times 10^{-4}$$

$$= 0.00025$$

frequency = 2.5×10^{-4} Hz

If the learner had not shown the working, this response would have scored 0 marks, as the final answer is incorrect. As the working was shown, marks for

substitution and evaluation (with error carried forward from the incorrect conversion) could be awarded.

Question 2

Q2(a) – learners were provided with a diagram representing a harmonic and asked to identify it from a choice of four options.

- 2 Figure 3 shows an incident wave and its reflected wave when a guitar string is plucked.

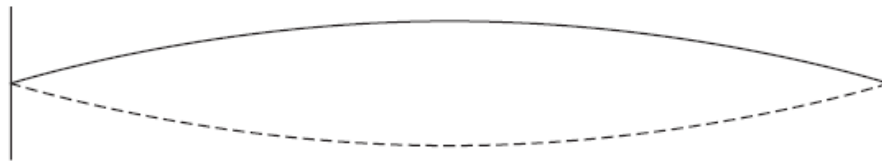


Figure 3

- (a) Identify the harmonic shown in Figure 3.

(1)

- ☐ **A** first
- ☐ **B** second
- ☐ **C** third
- ☐ **D** fourth

The majority of learners correctly chose option A, first.

Q2(b)(i) – this was a fairly challenging question for many learners, with the majority only able to gain 1 mark out of the 3 available, usually for correctly stating that increasing the tension in a guitar string would increase the frequency of the wave on the string. Very few learners were able to link ideas about tension, wave speed and frequency in order to explain why the wave frequency would increase. The response below gained full marks.

- (i) Explain how increasing the tension changes the frequency of the wave on the guitar string.

(3)

As the tension increases the wave speed increases, tension is directly proportional to the speed. Then, as the speed increases the frequency increases, the wave speed is directly proportional to the frequency. $V = 2 \times F$ $F \propto \frac{V}{2}$ So as the tension increases the frequency increases.

Q2(b)(ii) – many learners were able to use the equation provided in the question to perform some steps of the calculation correctly, gaining at least 2 marks. An example gaining full marks is shown below.

Show your working.

$$158 = \sqrt{\frac{T}{2.35 \times 10^{-3}}}$$

$$158^2 \times 2.35 \times 10^{-3}$$

$$= 58.7$$

tension = 58.7 N

Some learners struggled with rearranging the equation which contained a square root term but were still able to gain marks for process if the working was shown. An example of a response which gained 3 marks is shown below.

$$158 = \sqrt{\frac{T}{2.35 \times 10^{-3}}} = \frac{158}{2.35 \times 10^{-3}} = 67234.04$$

$$158 \times 2.35 \times 10^{-3} = 0.3713$$

$$0.3713 \times 158 = 58.61$$

$$\text{tension} = 0.3713 \text{ N}$$

The working in this response clearly shows how the value on the answer line has been derived and so marks could be awarded for correct substitution, rearrangement and evaluation (error carried forward). The only mark lost was that for dealing with the square root term. Again, centres are encouraged to ensure learners understand the importance of showing their working in calculations.

Q2(c) – few learners were able to explain how a standing wave is produced from an incident wave and its reflected wave. Many learners said the incident wave was reflected, which was not creditworthy since this information was given in the question stem. The following response gained full marks.

Figure 4 shows another incident wave and its reflected wave when the guitar string is plucked.

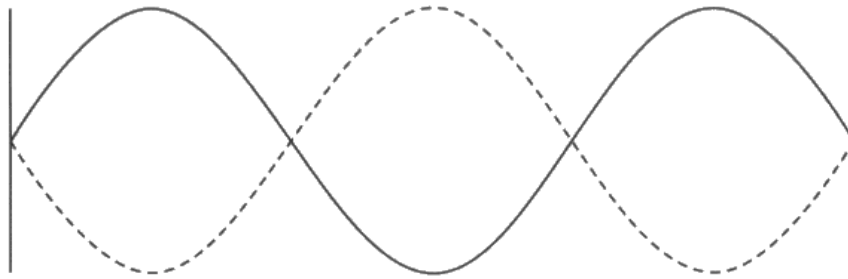


Figure 4

(c) Describe how these waves produce a standing wave.

(2)

both waves travel in opposite directions
and reflect back off the ends, forming nodes
and antinodes

Question 3

Q3(a)(i) – most learners were able to correctly state that gamma waves have the highest frequency in the electromagnetic spectrum as shown in the response below.

3 (a) Waves in the electromagnetic spectrum have many applications.

(i) State the region of the electromagnetic spectrum that has waves with the highest frequency.

(1)

Gamma waves

Some learners stated regions to the 'left' or 'right' which is insufficient for the mark as it is meaningless without a diagram of reference.

Q3(a)(ii) – many learners struggled to explain a feature of Bluetooth technology that reduces interference from other devices that use radio waves. A number of learners correctly identified frequency hopping as one such feature but did not go on to explain how this reduces interference. It is recommended that centres ensure learners understand the meaning of the command word 'explain' in order for them to access the highest marks available for a question. The following response gained full marks.

- (ii) Explain **one** feature of Bluetooth® technology that reduces interference from other devices that use radio waves.

(2)

Bluetooth devices use ~~a narrow~~ channel hopping, wherein they very quickly change the frequency of the emitted radio waves to avoid other devices using that specific frequency of radio wave, hence reducing interference.

Many learners suggested that the short range of Bluetooth would reduce the probability of interference from other devices, as shown in the response below. This was awarded 1 mark.

- (ii) Explain **one** feature of Bluetooth® technology that reduces interference from other devices that use radio waves.

(2)

Bluetooth ~~uses~~ emits short distance radio waves. Allowing for only a few devices in close proximity being able to receive the waves and connect.

Q3(b) – this question proved very challenging for many learners, with many misconceptions about optical fibre design, in particular, the role of the cladding. Many learners suggested the cladding protects a fragile glass core, while some said the cladding blocks out light. A number of responses did not address the question but simply gave generic information about optical fibres. The following response demonstrated excellent understanding of both optical fibre design and the application of this understanding to the answering the question.

(b) Explain how the optical fibre is designed for the total internal reflection of the pulse of light.

(3)

The cladding layer has a lower refractive index which makes it less dense, optically. The design ensures light pulses traveling from the core hit the boundary at an angle of incidence greater than the critical angle. As a result

(Total for Question 3 = 6 marks)

the light cannot refract out and is completely reflected back into the core allowing the pulse of light.

Question 4

Q4(a) – many learners struggled to describe how a microwave signal is transmitted from one mobile phone to another. Some learners tried to use information from the diagram in question 4(b) and talked about the use of satellites and radio waves. A common misconception was that the signal had a large enough wavelength to be able to reach directly from one mobile phone to another. The following response scored full marks.

4 Mobile phone networks use microwave signals.

(a) Describe how a signal is transmitted from one mobile phone to another.

(2)

The first phone transmits a signal to a signal tower and the tower ~~see~~ transmits the signal to the second phone which receives the signal.

Q4(b) – learners were provided with the diagram shown below.

Satellite communication uses radio waves and microwaves to transmit information.

Figure 6 shows how radio waves and microwaves are affected by the ionosphere.

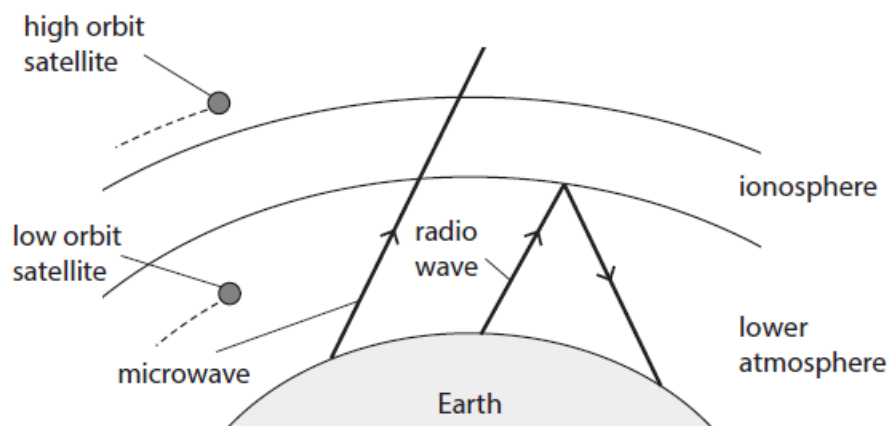


Figure 6

- (b) Discuss the differences between satellite communication using radio waves and satellite communication using microwaves.

Use information from Figure 6 in your answer.

(6)

It was pleasing to note that most learners used the information provided to help structure their response, but only a minority were able to demonstrate their knowledge and understanding of radio wave and microwave satellite communication with this.

The following response met the level 1 descriptors and was awarded 2 marks. The learner has made several statements using information from the diagram, but has been unable to link them, thus this response cannot achieve marks in level 2.

(6)

RA: Radiowaves will transmit the signals up and eventually bounce back down to Earth as it hit's the ionosphere. While microwave penetrates through the ionosphere

The next response contains a number of links made and was thus placed into level 2 and awarded 4 marks. This response includes some information not taken from the diagram: 'microwaves wavelength is shorter than radiowaves'. There are a

series of statements taken from the diagram: 'microwaves can penetrate the ionosphere', 'radiowaves are reflected by the ionosphere'.

There are some basic links present, for example, 'radiowaves are reflected by the ionosphere and only work in the lower atmosphere', 'radiowaves can only reach low orbit satellites', 'microwaves wavelength is shorter than radiowaves'. Although there is inclusion of some knowledge not taken from the diagram, it is **not** linked to the rest of the response and so does not take this response into level 3.

(b) Discuss the differences between satellite communication using radio waves and satellite communication using microwaves.

Use information from Figure 6 in your answer.

(6)

- Radio waves have a long wavelength
- Microwaves wavelength is shorter than Radiowaves
- Radiowaves are reflected by the ionosphere and only work in the lower atmosphere
- Microwaves can penetrate the ionosphere
- Radio waves can ~~only~~ only reach low orbit Satellites
- Microwaves can reach high orbit Satellites

The following response includes several points which are taken from the diagram in the question. There is also evidence of extra knowledge which is linked to the discussion. Examples include 'radiowaves have a smaller frequency than microwaves, which means they have a smaller bandwidth and carry less information', 'microwaves are affected by weather such as rain and fog meaning microwaves cannot carry information in harsh weather conditions'. The presence of these linkages plus the extra knowledge in the response placed it into level 3. There are a couple of statements which are incorrect: 'microwaves can only work in line of sight' and 'radiowaves can suffer interference, whereas microwaves do not'. As these statements did not contradict the rest of the response they were ignored, and marks were awarded for creditworthy content. This response was placed into level 3 and was awarded 6 marks.

Radiowaves have a smaller frequency than microwaves, which means they have a smaller bandwidth and carry less information.

Microwaves have a higher frequency than radiowaves, which means they have a higher bandwidth[↑]. Microwaves can carry more information than radiowaves.

Radiowaves have a larger wavelength so they can diffract from larger objects. ~~no~~ Microwaves have a shorter wavelength and cannot diffract large objects.

Microwaves can only work in line of sight, whereas Radiowaves can work out of sight, ^{*}

Microwaves are affected by weather such as rain and fog meaning microwaves cannot carry information in harsh weather conditions and information can be lost. Radiowaves are not affected by weather and can be used as under water communication.

Radiowaves can suffer interference, whereas microwaves do not.

Microwaves can pass through the ionosphere for better satellite communication, whereas radiowaves only reflect off the ionosphere.

^{*} This is due to them having a larger wavelength and being able ~~the~~ to diffract larger objects.

— Radiowaves can only reach the low orbiting satellites whereas microwaves can reach the high orbiting satellites. This is because microwaves ~~can~~ can pass through the ionosphere.

Summary

Biology

Across the paper, learners demonstrated strengths in recalling fundamental biological facts and interpreting straightforward diagrams, but performance was more variable when questions required application, mechanistic explanation, or sustained comparative reasoning. The assessment this series covered a wide range of specification content, including cell structure and function, staining techniques, microbial processes, physiological mechanisms such as synaptic transmission and gas exchange, interpretation of graphical data, and classification challenges in organisms such as *Euglena*.

Based on their performance on this paper, learners should:

- strengthen their ability to explain biological mechanisms rather than simply describe observable features or restate information from the question stem
- practice interpreting data and figures, ensuring they can extract key information and apply logical biological reasoning rather than offering narrative descriptions
- develop greater precision in scientific terminology, particularly when referring to organelles, cell types, and physiological processes
- consolidate understanding of commonly confused topics—such as differences between prokaryotes and eukaryotes, respiratory versus cardiovascular conditions, and the role of organelles in different cell types
- improve exam technique by reading questions carefully, responding directly to command words, and avoiding the inclusion of irrelevant or incorrect information that detracts from the quality of extended responses.

Chemistry

Based on their performance on this paper, learners should:

- ensure that they know what is meant by key terms such as property and use and formula and equation and can distinguish between the two
- ensure that they show their working in calculation questions so that if they do not achieve the correct final answer, they can be awarded part marks
- ensure that they are familiar with, and practice past paper examples of, the calculations in the specification such as moles calculations and using Avogadro's constant
- practice exam technique, ensuring that when answering a question, they relate their answer to the context given in the question rather than only giving generic answers, ensuring that they do not simply repeat information from the stem of the question
- ensure that they are familiar with the formula of common compounds and diatomic elements in the specification and be able to write these formulae using

the correct scientific conventions in terms of cases of letters and subscripts and removal of charges on ions where appropriate

- ensure that they not only can write state symbols, but understand when a substance is a solid, a liquid, a gas or aqueous
- ensure that they are familiar with the different types of substances in the specification and that they can explain their properties in terms of their bonding.

Physics

Based on their performance on this paper, learners should:

- check the units provided in a calculation question in order to identify if conversion into the SI unit is required
- practice rearranging equations involving a square root term
- be able to use the relationships in an equation to explain wave behaviour
- understand the requirements of the command word 'explain'
- understand how the design of an optical fibre is linked to total internal reflection of a pulse of light, in particular the nature and role of the cladding around the core
- learn how signals are transmitted between mobile phones.

Please refer to the Applied Science Specification, Additional Guidance and Sample Assessment Materials (SAMs) on the BTEC Nationals Applied Science website, located [here](#).